

Brutal Paper B62 - Reproduction in Plants | Simple Equations | Winds, Storms & Cyclones | Rational Numbers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A mango orchard has 480 bisexual flowers; only 35% of them are successfully pollinated and go on to form fruit. How many flowers fail to form fruit? [\[Reproduction in Plants\]](#)
 - (A) 312 flowers
 - (B) 465 flowers
 - (C) 320 flowers
 - (D) 168 flowers
 - (E) 156 flowers
 - (F) 445 flowers
 - (G) 300 flowers

2. Three page numbers that follow one after another in a book add up to 72. What is the largest of these three page numbers? [\[Simple Equations\]](#)
 - (A) 24
 - (B) 23
 - (C) 21
 - (D) 27
 - (E) 26
 - (F) 25

3. Normal air pressure at a coastal town is 1010 hPa, but inside a cyclone's eye it falls by 6%. What is the pressure in the eye? [\[Winds, Storms & Cyclones\]](#)
 - (A) 944.6 hPa
 - (B) 949.4 hPa
 - (C) 950 hPa
 - (D) 940 hPa
 - (E) 1070.6 hPa
 - (F) 60.6 hPa
 - (G) 1004 hPa

4. Add the rational numbers $-\frac{3}{4}$ and $\frac{5}{6}$. What is the sum in lowest terms? [\[Rational Numbers\]](#)
 - (A) $\frac{11}{12}$
 - (B) $-\frac{1}{12}$
 - (C) $\frac{2}{10}$
 - (D) $-\frac{19}{12}$
 - (E) $\frac{19}{12}$
 - (F) $\frac{1}{12}$
 - (G) $\frac{1}{2}$

5. Spirogyra reproduces by fragmentation: each filament breaks to double the count every cycle. Starting with 5 filaments, how many filaments are there after 4 cycles? [\[Reproduction in Plants\]](#)
- (A) 20 filaments
 - (B) 9 filaments
 - (C) 45 filaments
 - (D) 80 filaments
 - (E) 625 filaments
 - (F) 160 filaments
6. A number is multiplied by 4 and then 7 is subtracted, giving 53. What is the number? [\[Simple Equations\]](#)
- (A) 16
 - (B) 15
 - (C) 60
 - (D) 13.5
 - (E) 11.5
 - (F) 14
7. A cyclone is moving steadily at 15 km/h toward a coast that is 270 km away. How many hours will it take to reach the coast? [\[Winds, Storms & Cyclones\]](#)
- (A) 18 hours
 - (B) 9 hours
 - (C) 20 hours
 - (D) 12 hours
 - (E) 255 hours
 - (F) 27 hours
 - (G) 15 hours
8. Which is the greater of the two rational numbers $-\frac{2}{3}$ and $-\frac{3}{5}$? [\[Rational Numbers\]](#)
- (A) they are equal
 - (B) cannot be decided
 - (C) $-\frac{2}{3}$
 - (D) $-\frac{5}{3}$
 - (E) $-\frac{3}{5}$
 - (F) 0
 - (G) -1
9. A potato tuber has 12 eyes used for vegetative propagation. If each eye has a 75% chance of sprouting into a new plant, how many new plants are expected? [\[Reproduction in Plants\]](#)
- (A) 16 plants
 - (B) 12 plants
 - (C) 6 plants
 - (D) 4 plants
 - (E) 3 plants
 - (F) 8 plants
 - (G) 9 plants

10. Five times a number equals the same number increased by 36. Find the number. [\[Simple Equations\]](#)

- (A) 12
- (B) 36
- (C) 7.2
- (D) 18
- (E) 8
- (F) 45
- (G) 9

11. On heating, 1 cubic metre of air weighing 1.2 kg expands to 1.5 cubic metres while its mass stays the same. What is its new density? [\[Winds, Storms & Cyclones\]](#)

- (A) 0.8 kg per cubic metre
- (B) 0.9 kg per cubic metre
- (C) 1.8 kg per cubic metre
- (D) 1.2 kg per cubic metre
- (E) 1.5 kg per cubic metre
- (F) 2.0 kg per cubic metre
- (G) 0.6 kg per cubic metre

12. Find the product of $-\frac{2}{5}$ and $\frac{15}{-8}$, giving the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $\frac{30}{40}$
- (B) $\frac{6}{40}$
- (C) $-\frac{3}{4}$
- (D) $-\frac{30}{40}$
- (E) $\frac{3}{4}$
- (F) $\frac{4}{3}$

13. A bread mould releases 2000 spores into the air, but only 0.5% of them land in suitable spots and germinate. How many spores germinate? [\[Reproduction in Plants\]](#)

- (A) 20 spores
- (B) 50 spores
- (C) 5 spores
- (D) 1000 spores
- (E) 10 spores
- (F) 4 spores
- (G) 100 spores

14. When one third of a number is increased by 8, the result is 20. What is the number? [\[Simple Equations\]](#)

- (A) 36
- (B) 12
- (C) 28
- (D) 24
- (E) 84
- (F) 4
- (G) 40

15. A storm's winds blow at 90 km/h. Express this wind speed in metres per second. [\[Winds, Storms & Cyclones\]](#)

- (A) 324 m/s
- (B) 25 m/s
- (C) 50 m/s
- (D) 32.4 m/s
- (E) 5 m/s
- (F) 18 m/s
- (G) 15 m/s

16. Divide $-7/9$ by $14/-27$ and express the result in lowest terms. [\[Rational Numbers\]](#)

- (A) $27/18$
- (B) $189/126$
- (C) $3/2$
- (D) $-3/2$
- (E) $9/14$
- (F) $-2/3$

17. In one flower the number of stamens is 3 times the number of carpels, and together these reproductive parts total 24. How many stamens does the flower have? [\[Reproduction in Plants\]](#)

- (A) 18 stamens
- (B) 9 stamens
- (C) 6 stamens
- (D) 8 stamens
- (E) 12 stamens
- (F) 24 stamens

18. A rectangle has a perimeter of 54 cm, and its length is 3 cm more than its width. Find the length. [\[Simple Equations\]](#)

- (A) 27 cm
- (B) 13.5 cm
- (C) 12 cm
- (D) 21 cm
- (E) 18 cm
- (F) 15 cm
- (G) 24 cm

19. A barometer reading falls steadily from 1008 hPa to 984 hPa over 6 hours. By how much does the pressure fall each hour? [\[Winds, Storms & Cyclones\]](#)

- (A) 3 hPa per hour
- (B) 4 hPa per hour
- (C) 12 hPa per hour
- (D) 2 hPa per hour
- (E) 8 hPa per hour
- (F) 24 hPa per hour

20. What is the additive inverse of the rational number $-5/-7$? [\[Rational Numbers\]](#)

- (A) $-7/5$
- (B) $7/5$
- (C) 0
- (D) $-5/7$
- (E) $-5/-7$
- (F) $5/-7$ is its inverse

21. Yeast multiplies by budding so its count doubles every hour. Beginning with 8 yeast cells, how many cells are present after 5 hours? [\[Reproduction in Plants\]](#)

- (A) 256 cells
- (B) 48 cells
- (C) 1024 cells
- (D) 512 cells
- (E) 40 cells
- (F) 80 cells

22. A father is now 4 times as old as his son, and in 5 years he will be only 3 times as old. How old is the son now? [\[Simple Equations\]](#)

- (A) 12 years
- (B) 5 years
- (C) 20 years
- (D) 40 years
- (E) 15 years
- (F) 8 years
- (G) 10 years

23. Due to uneven heating, a spot on the equator receives 480 units of heat while a polar spot gets only one quarter as much. How many more units does the equator receive? [\[Winds, Storms & Cyclones\]](#)

- (A) 300 units
- (B) 420 units
- (C) 120 units
- (D) 240 units
- (E) 480 units
- (F) 600 units
- (G) 360 units

24. What is the reciprocal of the rational number $-8/15$? [\[Rational Numbers\]](#)

- (A) $-7/15$
- (B) $8/15$
- (C) $15/-8$ is positive
- (D) $-8/15$
- (E) 1
- (F) $-15/8$
- (G) $15/8$

25. A plant bears 90 flowers in which the ratio of unisexual to bisexual flowers is 2:3. How many bisexual flowers are there? [\[Reproduction in Plants\]](#)

- (A) 45 flowers
- (B) 72 flowers
- (C) 30 flowers
- (D) 60 flowers
- (E) 18 flowers
- (F) 54 flowers
- (G) 36 flowers

26. Two numbers add up to 60, and one of them is twice the other. What is the larger number? [\[Simple Equations\]](#)

- (A) 30
- (B) 45
- (C) 40
- (D) 20
- (E) 36
- (F) 60
- (G) 15

27. During the day a sea breeze blows from sea to land at 8 km/h. How long does a sailboat take to drift 24 km on this breeze? [\[Winds, Storms & Cyclones\]](#)

- (A) 6 hours
- (B) 1.5 hours
- (C) 16 hours
- (D) 3 hours
- (E) 2 hours
- (F) 4 hours
- (G) 192 hours

28. Find a rational number that lies exactly halfway between $\frac{1}{4}$ and $\frac{1}{3}$. [\[Rational Numbers\]](#)

- (A) $\frac{7}{12}$
- (B) $\frac{2}{7}$
- (C) $\frac{7}{24}$
- (D) $\frac{1}{7}$
- (E) $\frac{1}{3}$
- (F) $\frac{1}{24}$
- (G) $\frac{5}{12}$

29. A drumstick pod releases 15 winged seeds, and the wind carries 40% of them beyond 10 metres from the parent. How many seeds land beyond 10 metres? [\[Reproduction in Plants\]](#)

- (A) 5 seeds
- (B) 4 seeds
- (C) 6 seeds
- (D) 3 seeds
- (E) 9 seeds
- (F) 10 seeds
- (G) 8 seeds

- 30.** Adding 15 to twice a number gives the same result as taking 5 away from three times the number. Find the number. [\[Simple Equations\]](#)
- (A) 2
 - (B) 4
 - (C) 20
 - (D) 10
 - (E) -20
 - (F) 5
 - (G) 25
- 31.** Reduced pressure during a storm pulls up on a flat roof of area 20 square metres with a pressure difference of 300 newtons per square metre. What is the total upward force? [\[Winds, Storms & Cyclones\]](#)
- (A) 6000 N
 - (B) 15 N
 - (C) 3000 N
 - (D) 6 N
 - (E) 600 N
 - (F) 280 N
 - (G) 320 N
- 32.** Simplify the expression $\frac{3}{4} - (-\frac{5}{6}) + (-\frac{2}{3})$, giving the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $\frac{11}{12}$
 - (B) $\frac{7}{12}$
 - (C) $\frac{2}{3}$
 - (D) $\frac{13}{12}$
 - (E) $\frac{1}{12}$
 - (F) $-\frac{11}{12}$
 - (G) $\frac{27}{12}$
- 33.** An anther produces 1200 pollen grains, but only 1 in every 300 grains actually reaches a stigma. How many grains reach a stigma? [\[Reproduction in Plants\]](#)
- (A) 3 grains
 - (B) 40 grains
 - (C) 400 grains
 - (D) 4 grains
 - (E) 12 grains
 - (F) 2 grains
 - (G) 6 grains
- 34.** A quiz gives +4 marks for each correct answer and -1 for each wrong one. A student attempts all 50 questions and scores 120. How many did the student get correct? [\[Simple Equations\]](#)
- (A) 34
 - (B) 36
 - (C) 32
 - (D) 30
 - (E) 17
 - (F) 40
 - (G) 38

- 35.** A cyclone's wind speed strengthens from 60 km/h to 96 km/h. What is the percentage increase in wind speed? [\[Winds, Storms & Cyclones\]](#)
- (A) 36%
 - (B) 75%
 - (C) 160%
 - (D) 40%
 - (E) 96%
 - (F) 60%
- 36.** Express the rational number $-\frac{36}{48}$ in standard form (lowest terms with a positive denominator). [\[Rational Numbers\]](#)
- (A) $-\frac{9}{12}$
 - (B) $-\frac{4}{3}$
 - (C) $-\frac{3}{-4}$
 - (D) $-\frac{36}{48}$
 - (E) $\frac{3}{4}$
 - (F) $-\frac{6}{8}$
 - (G) $-\frac{3}{4}$
- 37.** A seed packet holds 250 seeds with a stated germination rate of 84%. How many seeds in the packet are expected to fail to germinate? [\[Reproduction in Plants\]](#)
- (A) 40 seeds
 - (B) 84 seeds
 - (C) 60 seeds
 - (D) 210 seeds
 - (E) 16 seeds
 - (F) 21 seeds
 - (G) 166 seeds
- 38.** Three quarters of a number is 9 less than the number itself. What is the number? [\[Simple Equations\]](#)
- (A) 24
 - (B) 27
 - (C) 36
 - (D) 12
 - (E) 18
 - (F) 48
- 39.** A thunderstorm cloud builds upward at a steady 8 metres per second. How high does it rise in 2 minutes? [\[Winds, Storms & Cyclones\]](#)
- (A) 16 m
 - (B) 480 m
 - (C) 240 m
 - (D) 128 m
 - (E) 120 m
 - (F) 1920 m
 - (G) 960 m

40. What do you get when you multiply $-\frac{4}{9}$ by its own reciprocal? [\[Rational Numbers\]](#)

- (A) 1
- (B) $\frac{81}{16}$
- (C) 0
- (D) $\frac{16}{81}$
- (E) -1
- (F) $-\frac{4}{9}$

41. Each pistil holds an ovary with 8 ovules. In a row of 25 such flowers that are all fully fertilized, how many seeds form in total? [\[Reproduction in Plants\]](#)

- (A) 160 seeds
- (B) 25 seeds
- (C) 8 seeds
- (D) 250 seeds
- (E) 320 seeds
- (F) 200 seeds
- (G) 33 seeds

42. A 26 m rope is cut into two pieces so that one piece is 4 m longer than the other. How long is the longer piece? [\[Simple Equations\]](#)

- (A) 17 m
- (B) 11 m
- (C) 18 m
- (D) 15 m
- (E) 22 m
- (F) 13 m
- (G) 9 m

43. During a cyclone 18 cm of rain falls in 4 hours at a steady rate. How much rain would fall in 10 hours at the same rate? [\[Winds, Storms & Cyclones\]](#)

- (A) 45 cm
- (B) 36 cm
- (C) 4.5 cm
- (D) 180 cm
- (E) 30 cm
- (F) 40 cm
- (G) 72 cm

44. What rational number should be added to $-\frac{5}{8}$ to give a result of $\frac{1}{4}$? [\[Rational Numbers\]](#)

- (A) $\frac{3}{8}$
- (B) $\frac{7}{8}$
- (C) $\frac{1}{8}$
- (D) $\frac{6}{8}$
- (E) $-\frac{7}{8}$
- (F) $-\frac{3}{8}$
- (G) $\frac{5}{8}$

45. A bee visits 18 flowers each minute for 25 minutes, and one third of these visits result in successful pollination. How many flowers are pollinated? [\[Reproduction in Plants\]](#)

- (A) 135 flowers
- (B) 300 flowers
- (C) 75 flowers
- (D) 225 flowers
- (E) 150 flowers
- (F) 450 flowers

46. Three pens costing Rs 20 each and 2 identical pencils together cost Rs 78. Find the cost of one pencil.

[\[Simple Equations\]](#)

- (A) Rs 15
- (B) Rs 4.5
- (C) Rs 13
- (D) Rs 12
- (E) Rs 18
- (F) Rs 9
- (G) Rs 6

47. A 2-litre balloon of cool morning air is warmed until it expands by 25% by noon. What is its new volume? [\[Winds, Storms & Cyclones\]](#)

- (A) 2.5 L
- (B) 0.5 L
- (C) 2.2 L
- (D) 3 L
- (E) 2.25 L
- (F) 1.5 L

48. Which rational number is equal to the decimal -0.85? [\[Rational Numbers\]](#)

- (A) $-\frac{17}{2}$
- (B) $-\frac{17}{200}$
- (C) $-\frac{85}{10}$
- (D) $-\frac{85}{1000}$
- (E) $-\frac{17}{20}$
- (F) $-\frac{8}{5}$

49. A bryophyllum plant has 6 leaves, each carrying 9 buds on its margin, and two thirds of these buds grow into new plants. How many new plants form? [\[Reproduction in Plants\]](#)

- (A) 54 plants
- (B) 48 plants
- (C) 12 plants
- (D) 45 plants
- (E) 27 plants
- (F) 36 plants

50. Twice the sum of a number and 5 equals 38. What is the number? [\[Simple Equations\]](#)

- (A) 9
- (B) 24
- (C) 16.5
- (D) 11
- (E) 19
- (F) 14
- (G) 28

51. A weather station logs wind speeds of 40, 55, 70, and 35 km/h over four readings. What is the average wind speed? [\[Winds, Storms & Cyclones\]](#)

- (A) 55 km/h
- (B) 50 km/h
- (C) 70 km/h
- (D) 45 km/h
- (E) 35 km/h
- (F) 48 km/h

52. Work out $-7/12$ minus $-5/12$ and give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) -1
- (B) $1/6$
- (C) $-1/6$
- (D) $-7/12$
- (E) $-1/12$
- (F) $-2/24$
- (G) -12

53. A watermelon contains 320 seeds in all, and $5/8$ of them are viable enough to germinate. How many viable seeds are in the fruit? [\[Reproduction in Plants\]](#)

- (A) 200 seeds
- (B) 180 seeds
- (C) 256 seeds
- (D) 100 seeds
- (E) 120 seeds
- (F) 160 seeds
- (G) 40 seeds

54. At a cyclone relief shelter the number of women is 12 more than the number of men, and the total is 96 people. How many men are there? [\[Simple Equations\]](#)

- (A) 12
- (B) 42
- (C) 45
- (D) 48
- (E) 84
- (F) 54

- 55.** A cyclone warning is issued exactly 36 hours before the predicted landfall at 09:00 on the 20th. When was the warning issued? [\[Winds, Storms & Cyclones\]](#)
- (A) 21:00 on the 19th
 - (B) 09:00 on the 19th
 - (C) 21:00 on the 17th
 - (D) 12:00 on the 18th
 - (E) 09:00 on the 18th
 - (F) 03:00 on the 19th
 - (G) 21:00 on the 18th
- 56.** Among the rational numbers $-\frac{1}{2}$, $-\frac{2}{3}$, $-\frac{3}{4}$, and $-\frac{1}{4}$, which one is the smallest? [\[Rational Numbers\]](#)
- (A) $-\frac{2}{3}$
 - (B) $-\frac{1}{4}$
 - (C) -1
 - (D) $-\frac{3}{4}$
 - (E) 0
 - (F) $-\frac{2}{4}$
- 57.** An alga in a fertilized pond doubles its count by fragmentation every 3 days. Starting from 50 algae, how many are present after 12 days? [\[Reproduction in Plants\]](#)
- (A) 250 algae
 - (B) 200 algae
 - (C) 650 algae
 - (D) 1600 algae
 - (E) 800 algae
 - (F) 400 algae
 - (G) 600 algae
- 58.** When 7 is subtracted from a number and the result is then divided by 3, the answer is 5. Find the number. [\[Simple Equations\]](#)
- (A) 18
 - (B) 15
 - (C) 17
 - (D) 8
 - (E) 12
 - (F) 22
 - (G) 36
- 59.** Each cubic metre of moist monsoon air holds 24 g of water vapour. How many kilograms of vapour are held in a hall containing 500 cubic metres of this air? [\[Winds, Storms & Cyclones\]](#)
- (A) 6 kg
 - (B) 120 kg
 - (C) 12 kg
 - (D) 24 kg
 - (E) 500 kg
 - (F) 1200 kg
 - (G) 1.2 kg

60. Two thirds of an unknown rational number equals $-10/9$. What is the number? [\[Rational Numbers\]](#)

- (A) $-3/5$
- (B) $5/3$
- (C) $-2/3$
- (D) $-10/6$
- (E) $-5/3$
- (F) $-15/9$
- (G) $-20/27$

61. A maize cob has 14 rows with 40 grains in each row, and 90% of the grains are filled because they were fertilized. How many grains are filled? [\[Reproduction in Plants\]](#)

- (A) 504 grains
- (B) 450 grains
- (C) 460 grains
- (D) 510 grains
- (E) 540 grains
- (F) 560 grains
- (G) 56 grains

62. The three angles of a triangle are x , $2x$, and $3x$ degrees. What is the measure of the largest angle?

[\[Simple Equations\]](#)

- (A) 100 degrees
- (B) 180 degrees
- (C) 30 degrees
- (D) 90 degrees
- (E) 60 degrees
- (F) 120 degrees
- (G) 45 degrees

63. Air rushing past one face of a flag moves at 12 m/s while the air on the other face moves at 4 m/s. Express the difference in their speeds in kilometres per hour. [\[Winds, Storms & Cyclones\]](#)

- (A) 16 km/h
- (B) 20 km/h
- (C) 2.22 km/h
- (D) 40 km/h
- (E) 28.8 km/h
- (F) 8 km/h

64. Evaluate the sum $-1/2 + 1/3 + (-1/6)$ and write the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-2/3$
- (B) $-1/3$
- (C) -1
- (D) $1/3$
- (E) $-1/6$
- (F) 0
- (G) $1/6$

65. Of 24 coconuts that fall near a shore, $\frac{3}{8}$ of them are carried by ocean water to a new beach where they sprout. How many coconuts reach a new beach? [\[Reproduction in Plants\]](#)

- (A) 8 coconuts
- (B) 9 coconuts
- (C) 12 coconuts
- (D) 6 coconuts
- (E) 3 coconuts
- (F) 18 coconuts

66. A taxi charges a fixed Rs 15 plus Rs 8 for every kilometre travelled. If a ride cost Rs 95, how many kilometres was it? [\[Simple Equations\]](#)

- (A) 11 km
- (B) 13.75 km
- (C) 12 km
- (D) 8 km
- (E) 10 km
- (F) 9 km
- (G) 80 km

67. A cyclone travels 540 km toward land over a period of 1.5 days. What is its average speed in kilometres per hour? [\[Winds, Storms & Cyclones\]](#)

- (A) 9 km/h
- (B) 360 km/h
- (C) 30 km/h
- (D) 540 km/h
- (E) 15 km/h
- (F) 22.5 km/h
- (G) 12 km/h

68. The temperature drops by $\frac{2}{5}$ degree Celsius every hour. What is the total temperature change after three and a half hours? [\[Rational Numbers\]](#)

- (A) $-\frac{2}{5}$ degrees
- (B) $\frac{7}{5}$ degrees
- (C) $-\frac{7}{5}$ degrees
- (D) $-\frac{1}{5}$ degrees
- (E) $-\frac{7}{10}$ degrees
- (F) $-\frac{14}{5}$ degrees
- (G) $-\frac{4}{5}$ degrees

69. A sunflower head bears 1000 tiny florets; each fertilized floret yields one seed, but 7% of the florets are sterile. How many seeds does the head produce? [\[Reproduction in Plants\]](#)

- (A) 900 seeds
- (B) 70 seeds
- (C) 830 seeds
- (D) 907 seeds
- (E) 930 seeds
- (F) 993 seeds
- (G) 1000 seeds

70. If 6 is subtracted from 5 times a number, the result equals 4 times the number plus 9. Find the number.

[Simple Equations]

- (A) 12
- (B) 15
- (C) 3
- (D) -3
- (E) 5
- (F) 9

71. Two anemometers read wind speeds of 48 km/h and 72 km/h. The second reading is what fraction greater than the first? [Winds, Storms & Cyclones]

- (A) $\frac{1}{2}$
- (B) 24
- (C) $\frac{3}{4}$
- (D) $\frac{1}{3}$
- (E) $\frac{2}{3}$
- (F) $\frac{1}{4}$
- (G) $\frac{3}{2}$

72. Find the value of the product $-\frac{2}{3} \times \frac{9}{-4} \times -\frac{1}{3}$ in lowest terms. [Rational Numbers]

- (A) $\frac{3}{2}$
- (B) $-\frac{1}{3}$
- (C) $\frac{1}{2}$
- (D) $-\frac{1}{2}$
- (E) $-\frac{1}{6}$
- (F) $\frac{6}{12}$
- (G) $-\frac{3}{2}$

73. A pea plant has 35 pods, but 4 of them are empty; every non-empty pod explosively flings out 8 seeds. How many seeds are flung in total? [Reproduction in Plants]

- (A) 240 seeds
- (B) 224 seeds
- (C) 256 seeds
- (D) 288 seeds
- (E) 280 seeds
- (F) 31 seeds
- (G) 248 seeds

74. A wind-speed reading w km/h is doubled and then 18 is subtracted, giving 122. What was the reading w ? [Simple Equations]

- (A) 80 km/h
- (B) 52 km/h
- (C) 70 km/h
- (D) 64 km/h
- (E) 86 km/h
- (F) 140 km/h
- (G) 61 km/h

75. A rising parcel of hot air has a volume of 300 cubic metres; as it cools it shrinks by one fifth. What is its new volume? [\[Winds, Storms & Cyclones\]](#)

- (A) 240 cubic metres
- (B) 200 cubic metres
- (C) 250 cubic metres
- (D) 360 cubic metres
- (E) 60 cubic metres
- (F) 270 cubic metres
- (G) 288 cubic metres

76. How many pieces of rope, each $\frac{3}{8}$ metre long, can be cut from a rope that is $\frac{9}{2}$ metres long? [\[Rational Numbers\]](#)

- (A) 12 pieces
- (B) 24 pieces
- (C) 16 pieces
- (D) 6 pieces
- (E) 4 pieces
- (F) 9 pieces
- (G) 8 pieces

77. In a papaya orchard 3 out of every 7 trees are male and bear no fruit. Out of 210 trees in all, how many trees actually bear fruit? [\[Reproduction in Plants\]](#)

- (A) 60 trees
- (B) 210 trees
- (C) 90 trees
- (D) 120 trees
- (E) 150 trees
- (F) 30 trees
- (G) 105 trees

78. The sum of a number and one fifth of itself is 30. What is the number? [\[Simple Equations\]](#)

- (A) 15
- (B) 36
- (C) 6
- (D) 5
- (E) 18
- (F) 25

79. A storm surge raises the sea by 3.5 m on top of a normal high tide of 1.2 m. What is the total height of the water above the usual level? [\[Winds, Storms & Cyclones\]](#)

- (A) 4.2 m
- (B) 1.2 m
- (C) 4.5 m
- (D) 2.3 m
- (E) 3.5 m
- (F) 5.7 m
- (G) 4.7 m

80. The sum of two rational numbers is $\frac{4}{9}$. If one of them is $-\frac{5}{9}$, what is the other number? [\[Rational Numbers\]](#)
- (A) $-\frac{1}{9}$
 - (B) 0
 - (C) 2
 - (D) $\frac{1}{9}$
 - (E) -1
 - (F) 1
81. A flower starts with 800 viable pollen grains and loses one quarter of whatever remains each day. How many viable grains are left after 2 full days? [\[Reproduction in Plants\]](#)
- (A) 400 grains
 - (B) 350 grains
 - (C) 450 grains
 - (D) 600 grains
 - (E) 225 grains
 - (F) 500 grains
82. Two consecutive even numbers add up to 90. What is the smaller of the two numbers? [\[Simple Equations\]](#)
- (A) 42
 - (B) 44
 - (C) 45
 - (D) 46
 - (E) 88
 - (F) 22
 - (G) 48
83. A cyclone is 400 km across, while its calm central eye is 40 km across. The eye's diameter is what percent of the whole cyclone's diameter? [\[Winds, Storms & Cyclones\]](#)
- (A) 25%
 - (B) 10%
 - (C) 4%
 - (D) 90%
 - (E) 20%
 - (F) 40%
84. Evaluate $-\frac{3}{7}$ divided by $\frac{6}{14}$ and give the result in lowest terms. [\[Rational Numbers\]](#)
- (A) $-\frac{9}{49}$
 - (B) 1
 - (C) 2
 - (D) $\frac{1}{2}$
 - (E) $\frac{42}{42}$
 - (F) -1

- 85.** Five flowers each shed 240 pollen grains from their anthers, but one flower's anther fails to open. How many pollen grains are shed altogether? [\[Reproduction in Plants\]](#)
- (A) 800 grains
 - (B) 1200 grains
 - (C) 720 grains
 - (D) 480 grains
 - (E) 1000 grains
 - (F) 960 grains
 - (G) 240 grains
- 86.** A seedling grows 2 cm every week from its starting height, and after 6 weeks it stands 17 cm tall. What was its starting height? [\[Simple Equations\]](#)
- (A) 7 cm
 - (B) 17 cm
 - (C) 3 cm
 - (D) 12 cm
 - (E) 2 cm
 - (F) 29 cm
 - (G) 5 cm
- 87.** During a storm, thunder is heard 6 seconds after the lightning is seen. If sound travels at 340 m/s, how far away is the storm? [\[Winds, Storms & Cyclones\]](#)
- (A) 2400 m
 - (B) 1700 m
 - (C) 3400 m
 - (D) 2040 m
 - (E) 56.7 m
 - (F) 6 m
 - (G) 340 m
- 88.** Of the two rational numbers $\frac{5}{6}$ and $\frac{7}{9}$, which is larger and by how much? [\[Rational Numbers\]](#)
- (A) $\frac{7}{9}$ is larger by $\frac{1}{18}$ too
 - (B) $\frac{5}{6}$ is larger by $\frac{1}{18}$
 - (C) $\frac{7}{9}$ is larger by $\frac{2}{15}$
 - (D) $\frac{5}{6}$ is larger by $\frac{1}{3}$
 - (E) they are equal
 - (F) $\frac{7}{9}$ is larger by $\frac{1}{18}$
 - (G) $\frac{5}{6}$ is larger by $\frac{1}{9}$
- 89.** In a germination test, three trays each hold 60 seeds and germinate at 80%, 90%, and 70%. How many seeds germinate in total across the three trays? [\[Reproduction in Plants\]](#)
- (A) 120 seeds
 - (B) 150 seeds
 - (C) 162 seeds
 - (D) 144 seeds
 - (E) 36 seeds
 - (F) 135 seeds
 - (G) 180 seeds

90. The difference between a number and one third of itself is 16. Find the number. [\[Simple Equations\]](#)

- (A) 20
- (B) 12
- (C) 48
- (D) 32
- (E) 8
- (F) 24

91. Wind presses on a wall with a pressure of 250 newtons per square metre. If the wall is 8 m long and 2.5 m high, what total force does the wind exert? [\[Winds, Storms & Cyclones\]](#)

- (A) 260 N
- (B) 2500 N
- (C) 5000 N
- (D) 20 N
- (E) 500 N
- (F) 1250 N
- (G) 10000 N

92. On a number line, point P is at $-\frac{2}{3}$ and point Q is at $\frac{5}{6}$. What is the distance from P to Q? [\[Rational Numbers\]](#)

- (A) $\frac{3}{2}$
- (B) $\frac{9}{12}$
- (C) $-\frac{3}{2}$
- (D) $\frac{7}{6}$
- (E) $\frac{2}{3}$
- (F) $\frac{1}{2}$
- (G) $\frac{1}{6}$

93. A strawberry plant makes 4 daughter plants by runners each season, and every daughter makes 4 of its own the next season. Starting from 1 plant, how many brand-new plants appear in the second season?

[\[Reproduction in Plants\]](#)

- (A) 8 plants
- (B) 32 plants
- (C) 4 plants
- (D) 64 plants
- (E) 16 plants
- (F) 12 plants
- (G) 20 plants

94. Five times a number is as much above 30 as it is below 70. What is the number? [\[Simple Equations\]](#)

- (A) 2
- (B) 20
- (C) 8
- (D) 5
- (E) 7
- (F) 10

95. As a cyclone weakens, its wind speed of 120 km/h drops by 35%. What is the new wind speed? [\[Winds,](#)

[Storms & Cyclones\]](#)

- (A) 42 km/h
- (B) 156 km/h
- (C) 78 km/h
- (D) 72 km/h
- (E) 84 km/h
- (F) 90 km/h

96. By how much is $\frac{7}{10}$ greater than $\frac{2}{5}$? [\[Rational Numbers\]](#)

- (A) $\frac{1}{10}$
- (B) $\frac{9}{10}$
- (C) $\frac{1}{2}$
- (D) $\frac{2}{10}$
- (E) $\frac{3}{5}$
- (F) $\frac{3}{10}$
- (G) $\frac{5}{5}$

97. One flower has 5 sepals, 5 petals, 10 stamens, and 1 pistil. What fraction of all its floral parts are stamens? [\[Reproduction in Plants\]](#)

- (A) $\frac{5}{11}$
- (B) $\frac{2}{21}$
- (C) $\frac{10}{21}$
- (D) $\frac{5}{21}$
- (E) $\frac{10}{11}$
- (F) $\frac{1}{21}$

98. A square's perimeter is exactly equal to the perimeter of an equilateral triangle whose side is 16 cm.

Find the side of the square. [\[Simple Equations\]](#)

- (A) 4 cm
- (B) 12 cm
- (C) 6 cm
- (D) 24 cm
- (E) 16 cm
- (F) 8 cm

99. A region receives 120 cm of rain in a year, and three quarters of it arrives during the four monsoon months. How much rain falls in the monsoon season? [\[Winds, Storms & Cyclones\]](#)

- (A) 75 cm
- (B) 90 cm
- (C) 30 cm
- (D) 100 cm
- (E) 45 cm
- (F) 40 cm
- (G) 160 cm

100. Evaluate the expression $\frac{3}{5} \times (-\frac{10}{9}) + \frac{1}{3}$ and give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $\frac{2}{3}$
- (B) -1
- (C) $-\frac{1}{9}$
- (D) 0
- (E) $-\frac{1}{3}$
- (F) $\frac{1}{3}$
- (G) $-\frac{2}{3}$